



## Tutorial exercises (2)

### 1. EXERCISE

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Write an algorithm that gives values to two integer variables named **A** and **B**, and that displays their values, the product, and the sum, in this form :

**A=3 , B=5**

**A\*B=15**

**A+B=8**

### 2. EXERCISE

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Write an algorithm that requires two integers and provides their sum and product. the dialogue with the user will look like this :

Give two numbers : 45 3

Their sum is : 48

Their product is : 135

### 3. EXERCISE

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1. Write an algorithm that checks whether a number given by the user is even or odd.
2. Write an algorithm that displays the absolute value of a given number.
3. Write the algorithm that displays the next positive value if the value given by the user is a positive value, and the previous negative value if the value given by the user is a negative value.
4. Write an algorithm that reads three integers and finds whether or not they are in ascending order. That is, if  $a, b, c$  denote these numbers, they will have to check the mathematical inequality  $a < b < c$ .

### 4. EXERCISE

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Write an algorithm that exchanges the values of 2 real variables:

- Using an intermediate variable.
- Without using an intermediate variable.

## 5. EXERCISE

Insuranc offers three prices (**Green,Orange,Red**) depending on the age and number of accidents of drivers.

| Number of Accidents | under 25 years | 25 years and over |
|---------------------|----------------|-------------------|
| 0                   | orange         | green             |
| 1 or 2              | Red            | Orange            |
| 3 to 6              | Not insured    | Red               |
| 7 or over           | Not insured    | Not insured       |

- Write an algorithm that displays the rate after entering the age and number of accidents of a driver.

## 6. EXERCISE

We want to compare two telephone subscription offers.

| Offers    | Fixed  | Price per minute |
|-----------|--------|------------------|
| Telecom 1 | 200 DA | 2 DA             |
| Telecom 2 | 300 DA | 1.50 DA          |

The invoice is calculated with a fixed amount ( amount to be paid every month) and a part proportional to the time spent on the phone (indicated in minutes).

- Write an algorithm that indicates the most interesting operator after entering the average monthly consumption (in minutes).

## 7. EXERCISE

In the computer sciences department, we have 5 examination rooms: R1, R2, R3, R4 and R5, each with 40 places. The number of students in the 1<sup>st</sup> year is 180, and each student has a registration number.

1. Write an algorithm that asks for a student's number to direct them to an exam room.
2. Write the same algorithm using the **Case** instruction.

## 8. EXERCISE

Write an algorithm that reads a tax-exclusive price and calculates and displays the corresponding tax-inclusive price (with a VAT rate of 17%). It then establishes a discount whose rate is as follows:

- 0 % for an amount including tax of less than 1000 DA,
- 1 % for an amount including tax greater than or equal to 1000 DA and less than 2000 DA,
- 2 % for an amount including tax greater than or equal to 2000 DA and less than 5000 DA,
- 5 % for an amount including tax greater than or equal to 5000 DA.

The discount obtained and the new amount including tax will be displayed.

## 9. EXERCISE

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1. Write an algorithm which, starting from a given date, displays the number of remaining days before the end of the month.
2. Write an algorithm that asks the user for the number of a month in order to display the number of days in the following month.
3. Write an algorithm that asks the user for a date to display the date of the next day.
4. Write an algorithm that asks the user for a date to display the previous day's date.
5. Write an algorithm that converts the time format of an electronic clock from AM and PM to 24 hours.

## 10. EXERCISE

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In an oral examination, (3) candidates are examined one after the other by an examiner, who asks each candidate questions and then gives them a mark. The following table shows the examination time for each candidate:

| Candidate | Examination time          |
|-----------|---------------------------|
| C1        | 10 minutes and 52         |
| C2        | 9 minutes and 37 seconds  |
| C3        | 11 minutes and 24 seconds |

After each candidate has passed, the examiner takes **3 minutes** to give the mark and start with the next candidate.

- If the exam starts at time T (H:M:S), write the algorithm that displays the start and end times for each student.

## 11. EXERCISE

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A medical laboratory carries out analyses for patients. The result of each analysis is obtained after **15 days** from the date of analysis.

1. Write an algorithm that asks the user for the date of the analysis (day, month, year) to give the result date.
2. Write the previous algorithm using the Case instruction.